Supplementary Table S2. Mapping Relationships, Confidence Scores, Reasons for Non-Mapping, and Laterality Qualifiers in the AI-READI Survey-to-OMOP Mapping

A. Mapped data elements by SKOS mapping relationship and confidence score

Target concept type	SKOS mapping relationship	Confidence 1.0	Confidence 0.75	Confidence 0.5	Total
Standard OMOP concepts		**1,327**	**83**	**2**	**1,412**
	Exact match	1,171	3	0	1,174
	Close match	156	79	0	235
	Related match	0	1	2	3
AI-READI custom concepts		**488**	**0**	**0**	**488**
	Exact match	488	0	0	488
Total mapped elements		**1,815**	**83**	**2**	**1,900**

Counts are mapped data elements (n = 1,900). Mappings with an exact or close match and a confidence score of at least 0.75 were accepted; all others were flagged for working group review. Two related-match mappings at confidence 0.5 were retained as standard concepts after review. All AI-READI custom concept mappings are exact matches at confidence 1.0.

B. Data elements designated for mapping but not mapped to concepts, by reason

Reason	n	% of 104	Description
Laboratory reference ranges	46	44.2	Minimum and maximum reference values accompanying laboratory results; handled as measurement attributes (e.g., range_low/range_high) during ETL
REDCap survey timestamps	34	32.7	Survey start and completion timestamps; REDCap system metadata rather than survey content
Specimen handling fields	12	11.5	Specimen collection and processing dates and handling notes
BCVA data-entry modifiers	10	9.6	Best-corrected visual acuity "not done" indicators and sphere sign (+/−) modifiers applied during ETL
REDCap administrative field	1	1.0	Survey-scheduling workflow flag
MoCA paper-administration flag	1	1.0	Indicator that the MoCA was administered on paper
Total	**104**	**100.0**	

These 104 elements were designated for mapping but intentionally not assigned OMOP concepts; they were handled as attributes or metadata during ETL. Survey timestamps were assigned AI-READI custom concepts in the mapping file but are counted here because they are system metadata rather than survey content.

C. Data elements with laterality recorded as a qualifier

Assessment	Right (n)	Left (n)	Total	Qualifier concept
Retinal imaging acquisitions	66	66	132	Eye
Best-corrected visual acuity	6	6	12	Eye
Contrast sensitivity (photopic and mesopic)	6	6	12	Eye
Monofilament foot sensation (felt/tested totals)	2	2	4	Foot
Total	**80**	**80**	**160**	

Laterality was recorded with standard concepts: Right eye (45876703) and Left eye (45883829); Structure of right foot (4298982) and Structure of left foot (4320144). Site-level monofilament results additionally use AI-READI custom site qualifiers (concept IDs 2005200605–2005200624).

All counts were generated from the final mapping file using count_concept_map.sql (see manuscript-tvst2026/count_concept_map.sql). BCVA, best-corrected visual acuity; ETL, extract, transform, load; MoCA, Montreal Cognitive Assessment; SKOS, Simple Knowledge Organization System.